SGA 2: Application to the Fundamental Group

Bounded source-aligned working unit — 20 July 2026

Authority. Corrected French TeX lines 3446–3453; original printed p. 117, physical source-PDF p. 101, and recomposed running p. 93. These locator systems are kept distinct. Blank line 3454 is excluded; the raw cursor is line 3454 and the next substantive cursor is line 3455, the one-line derivation. The editorial marker (3) is retained, while its long note at line 3456 remains outside this statement-only unit. The external English candidate is comparison-only.

Corollary 2.6*[(3)]*. Suppose that $\mathrm{Lef}(X, Y)$ holds and that Y is connected. Then every open neighborhood U of Y is connected, and the natural homomorphism $\pi_1(Y) \longrightarrow \pi_1(U)$ is surjective. Suppose, moreover, that $\mathrm{Leff}(X, Y)$ holds. Then the natural homomorphism

$$\pi_1(Y) \longrightarrow \varprojlim_{U} \pi_1(U)$$

is an isomorphism. (N.B. A “base point” in Y is understood to have been chosen and is also taken as a base point in X for the definition of the fundamental groups.)